Tharani Kumar M

Hosur, India | tharanikumarpro@gmail.com | +91-7094054257
linkedin.com/in/tharanikumarm | github.com/tharanikumarm

Objective

Electrical and Electronics Engineer with experience in embedded system design, hardware prototyping, and embedded software development. Interested in Embedded Systems and Automation domains with hands-on exposure to AUTOSAR architecture, data acquisition systems and hardware validation.

Skills

- **Programming:** Embedded C, Python
- **Embedded Systems:** STM32, GPIO, ADC, I2C, SPI, UART
- **Architecture & Methodologies:** AUTOSAR classic,Finite State Machine,V-Model, FreeRTOS ,Version control(GIT)
- **Tools & Validation:** Cppcheck (Static Analysis), Oscilloscope, Logic Analyzer
- **Hardware Design:** KiCad, Proteus, PCB Prototyping

Projects

Linear Actuation System for Transducer Testing Machine

- Designed a ball screw based linear actuation system for transducer testing and selected the stepper motor using torque calculations.
- Developed embedded UI using keypad and 16×2 LCD.
- Implemented data acquisition using AD7606 ADC and SD card logging with RTC timestamping.
- Developed software using AUTOSAR architecture and MISRA-C guidelines.
- Performed static analysis, hardware assembly, soldering, and system validation.

Automated Coil Winding Machine

- Designed and developed an automated coil winding machine capable of winding 0.07 mm copper wire by integrating the mechanical and electrical systems, including torque-based stepper motor selection and machine architecture.
- Developed embedded firmware on the STM32F446RE using Embedded C in STM32CubeIDE with a combination of HAL and Bare-Metal programming, implementing a USB keyboard-based user interface, 20×4 I^2C LCD display, and proximity sensor-based position control while following AUTOSAR architecture and MISRA-C coding guidelines.
- Automated the manual coil winding process, reducing manufacturing cost by approximately **86.67%** while improving production efficiency, repeatability, and operator usability.

Experience

Loyal Wingman Technologies Private Limited — Jun 2025 – Present
Embedded Engineer

- Developed and validated embedded firmware.
- Participated in hardware bring-up and testing.
- Performed debugging using oscilloscopes and logic analyzers.

Education

Bachelor of Engineering in Electrical and Electronics Engineering — 2021 – 2025
Kongu Engineering College
Tamil Nadu, India
CGPA: 7.44 / 10